EMG / ECG Language Processor — wiring

Parts: 1x Arduino Uno, 2x MyoWare 2.0, 1x AD8232 module,
9x Ag/AgCl snap electrodes (single-use), jumper wires, USB cable.
No extra components: each sensor board does its own analog filtering;
the Uno does the digital side (threshold, debounce, bits -> ASCII, heart rate).

WIRING LIST

Sensor pin -> Uno pin
MyoWare LEFT +Vs/GND -> 5V / GND
MyoWare LEFT ENV -> A0 (left wrist = bit 0)
MyoWare RIGHT +Vs/GND -> 5V / GND
MyoWare RIGHT ENV -> A1 (right wrist = bit 1)
AD8232 3.3V / GND -> 3.3V / GND
AD8232 OUTPUT -> A2 (heartbeat waveform)
AD8232 LO+ / LO- -> A3 / A4 (lead-off = electrode fell off)

WHY THE FOREARM, NOT THE WRIST ITSELF

The muscles that move the wrist up/down live in the forearm:
the wrist is mostly tendon and gives a weak, noisy signal.
Flexing the wrist down fires the palm-side flexors, so each
MyoWare sits on the flexors of one arm:
flex LEFT = 0, flex RIGHT = 1, flex BOTH = commit letter.
ECG electrodes DO go on the wrists (lead I: right vs left arm).

PWR_FLAG = KiCad ERC marker only (not a part).

MyoWare 2.0: MID + END snap onto the board itself (board sits on the muscle):
REF uses the short cable. ENV output = smoothed muscle activity (on-board filtering).

+5V
M1
MyoWare 2.0 — LEFT forearm
+Vs
ENV
RAW
RECT
GND
MID
END
REF
E1
E2
E3

MID: middle of the wrist-flexor muscle belly,
upper third of the forearm (palm side)
END: 1-2 cm toward the wrist, in line with the muscle
REF: bony spot — back of elbow or wrist bone

+5V
M2
MyoWare 2.0 — RIGHT forearm
+Vs
ENV
RAW
RECT
GND
MID
END
REF
E4
E5
E6

MID: middle of the wrist-flexor muscle belly,
upper third of the forearm (palm side)
END: 1-2 cm toward the wrist, in line with the muscle
REF: bony spot — back of elbow or wrist bone

+3V3
+5V
VIN
3V3
+5V
D0/RX
D1/TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
RESET
IOREF
AREF
A0
A1
A2
A3
SDA/A4
SCL/A5
SDA/A4
SCL/A5
GND
GND
GND
A1
Arduino Uno R3

USB cable to laptop:
power in + serial text out.
Laptop does text-to-speech
and plays the alarm.

A3 / A4 read as digital
inputs (digitalRead).

PWR_FLAG
GND

+3V3
M3
AD8232 heart monitor module
3.3V
OUTPUT
LO+
LO-
SDN
GND
RA
LA
RL
E7
E8
E9

RA (red): inside of RIGHT wrist
LA (yellow): inside of LEFT wrist
RL (green): right forearm (reference)

Leads plug into the module's 3.5 mm jack.
Colors = standard SparkFun lead cable.

AD8232 runs on 3.3 V ONLY (never 5 V).
All ECG filtering is on the module.
SDN left open (pulled up on board).

SAFETY: laptop on battery (unplugged from the wall) while electrodes are on a person.
Heart-rate alarm is a warning aid, not a medical device.

Uno + 2x MyoWare 2.0 + AD8232 module — breakout boards, no custom PCB
Proof of concept — USB powered from a laptop on battery, NOT galvanically isolated

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Title: EMG/ECG Language Processor

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